Malaria



WHAT IS MALARIA ?

A febrile illness caused by a parasite transmitted by mosquitoes.

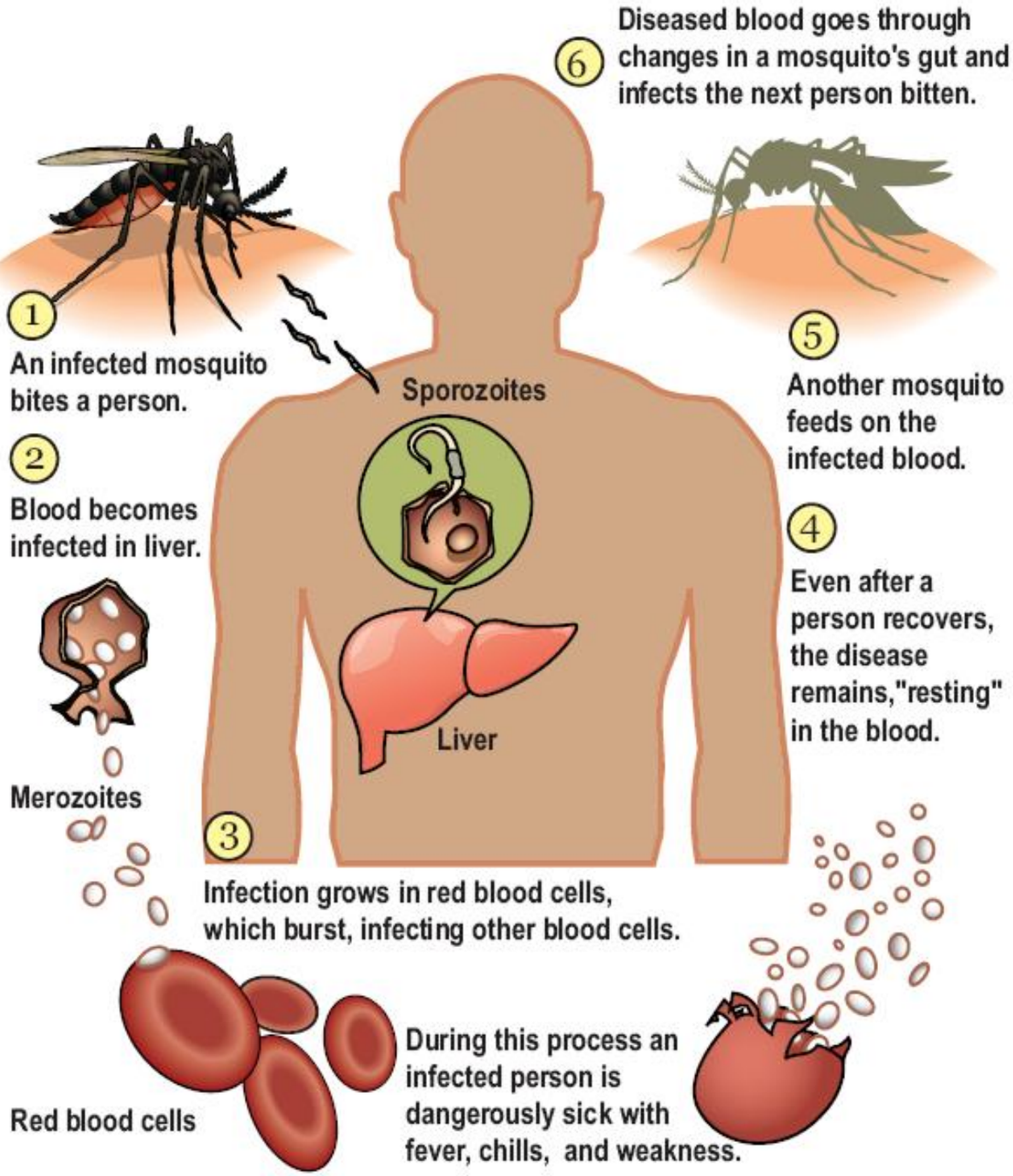
There are four types of Plasmodium parasites:

- **FALCIPARUM -** *Very common in West Africa*
  Untreated, it can kill you

- **Vivax** causes milder disease and unless correctly treated can hide in the liver causing malaria relapses many months and even years later

- Plasmodium **Malariae and Ovale** are less important species in West Africa

What happens when you get bitten by an infected mosquito?



Malaria Prevention
Saves Lives

If diagnosed and treated correctly
and without delay, recovery is complete

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Clinical presentation

Acute febrile illness, may have periodic febrile paroxysms every 48 – 72 hours with Afebrile asymptomatic intervals
Tendency to recrudesce or relapse over months to years
Anemia, thrombocytopenia, jaundice, hepatosplenomegaly, respiratory distress syndrome, renal dysfunction, hypoglycemia, mental status changes, tropical splenomegaly syndrome

Clinical presentation

Early symptoms
- Headache
- Malaise
- Fatigue
- Nausea
- Muscular pains
- Slight diarrhea
- Slight fever, usually not intermittent

Could mistake for influenza or gastrointestinal infection





Malaria risk awareness

Cerebral malaria is a consequence of severe malaria with supplementary signs:

- Irritability
- Confusion
- Convulsions
- Coma

Clinical presentation

Signs

- Anemia
- Thrombocytopenia
- Jaundice
- Hepatosplenomegaly
- respiratory distress syndrome
- renal dysfunction
- Hypoglycemia
- Mental status changes
- Tropical splenomegaly syndrome

Complications

Acute Renal Failure

Acute Pulmonary Edema

(Acute Respiratory Distress Syndrome)

Hypoglycemia

Spontaneous Bleeding

Convulsions

Aspiration pneumonia

Bacterial Sepsis

Laboratory Diagnosis

LABORATORY DIAGNOSIS OF MALARIA IS CONFIRMED BY
THE DEMONSTRATION OF MALARIA PARASITE IN
THE BLOOD FILM UNDER MICROSCOPE EXAMINATION

- **THIN FILM**
- **THICK FILM**



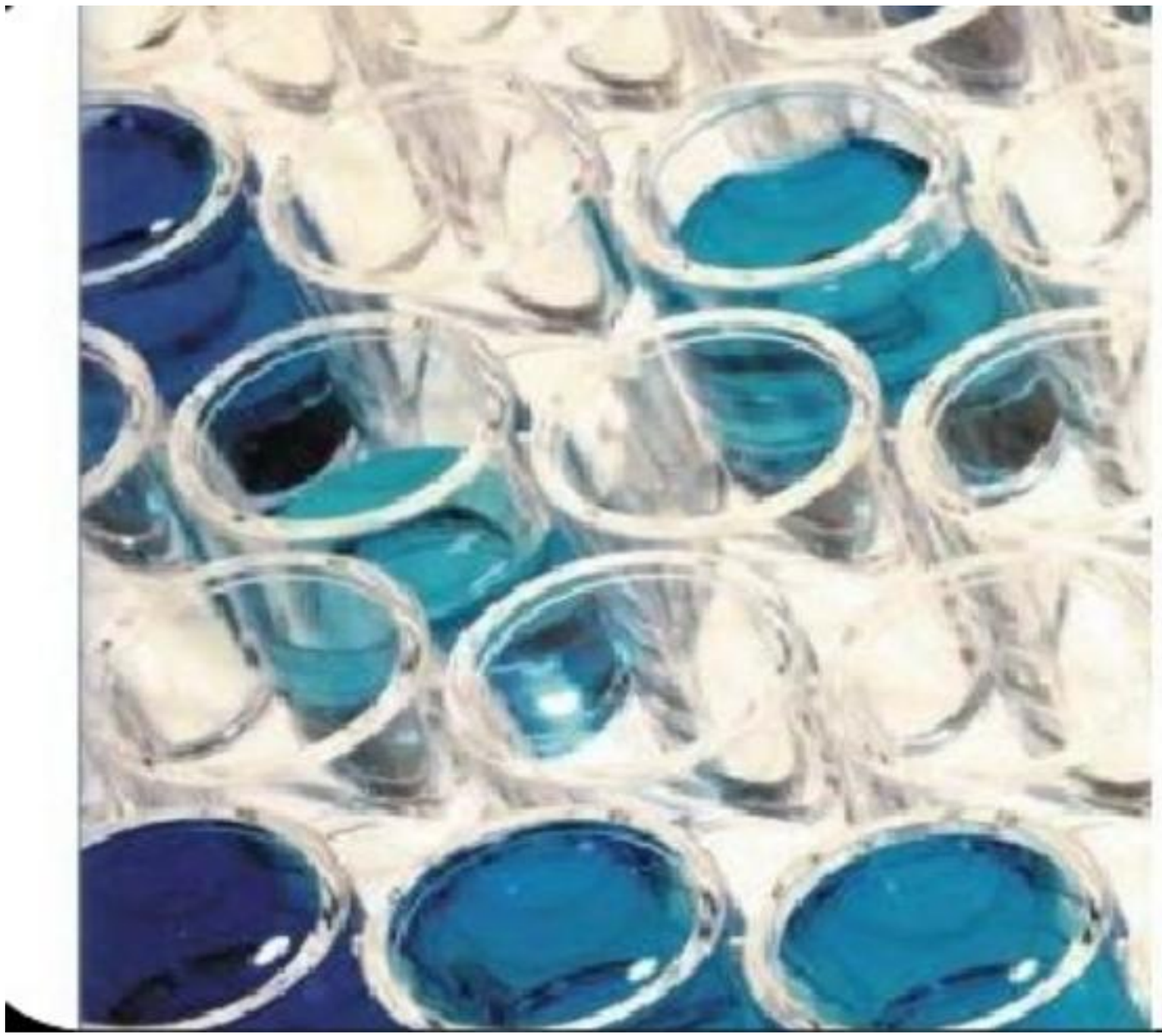
Antigen detection methods

Various test kits are available to detect antigens derived from malaria parasites . Such immunologic ("immunochromatographic") tests most often use a dipstick or cassette format, and provide results in 2-15 minutes. These "Rapid Diagnostic Tests" (RDTs) offer a useful alternative to microscopy in situations where reliable microscopic diagnosis is not available

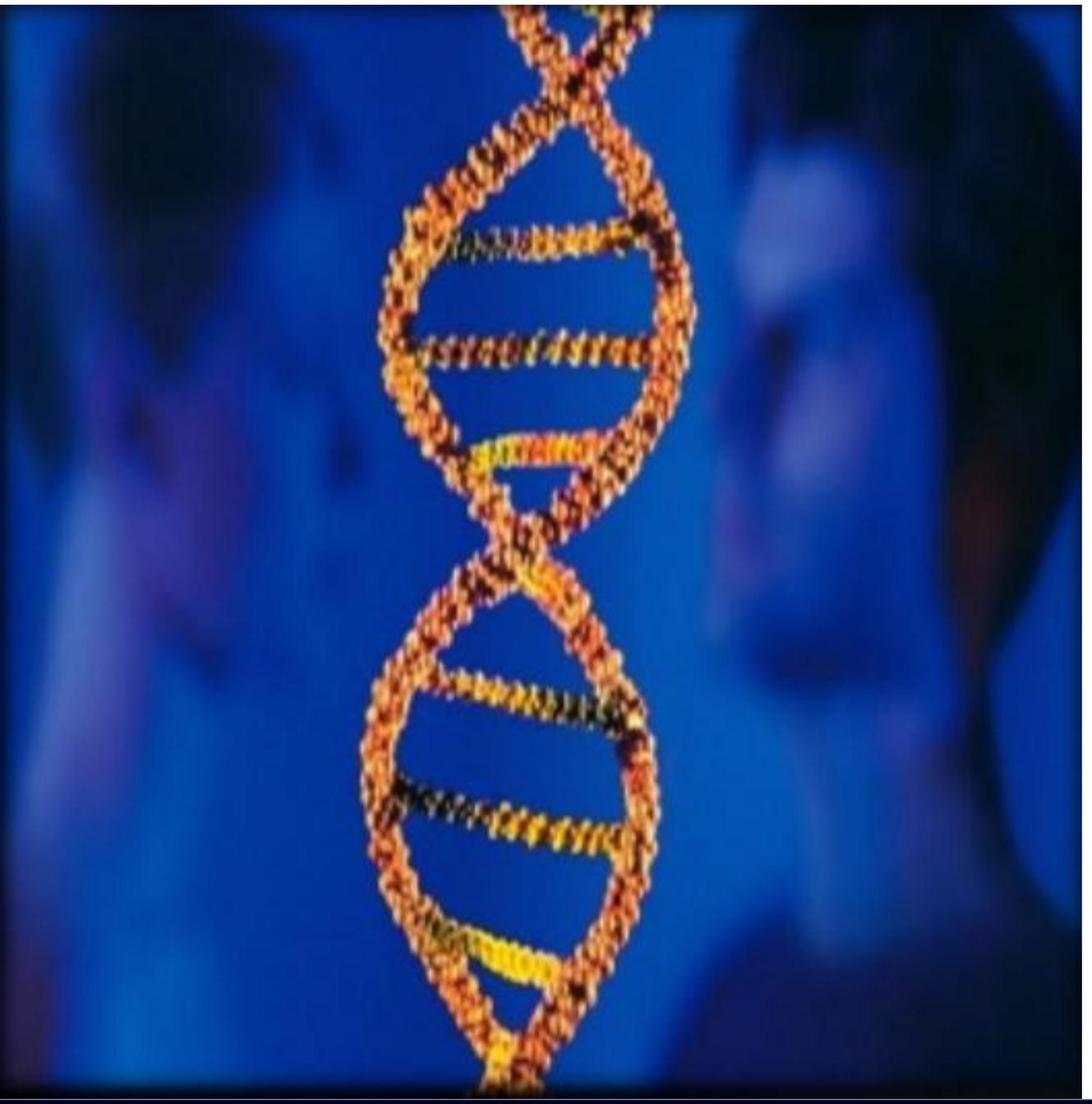


SEROLOGY IN MALARIA

Serology detects antibodies against malaria parasites , using either indirect immunofluorescence (IFA) or enzyme-linked immunosorbent assay (ELISA). Serology does not detect current infection but rather measures past exposure.



MOLECULAR DIAGNOSIS OF MALARIA



Parasite nucleic acids are detected using polymerase chain reaction (PCR). Although this technique may be slightly more sensitive than smear microscopy, it is of limited utility for the diagnosis of acutely ill patients in the standard healthcare setting. PCR results are often not available quickly enough to be of value in establishing the diagnosis of malaria infection

PCR IS USEFUL IN SPECIFIC DETECTION

PCR is most useful for confirming the species of malarial parasite after the diagnosis has been established by either smear microscopy or RDT



Control



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Vector Control

- Drainage and filling
- Chemical and biological larvicides
- Indoor residual insecticide spraying
- Outdoor residual insecticide spraying

costly and effective

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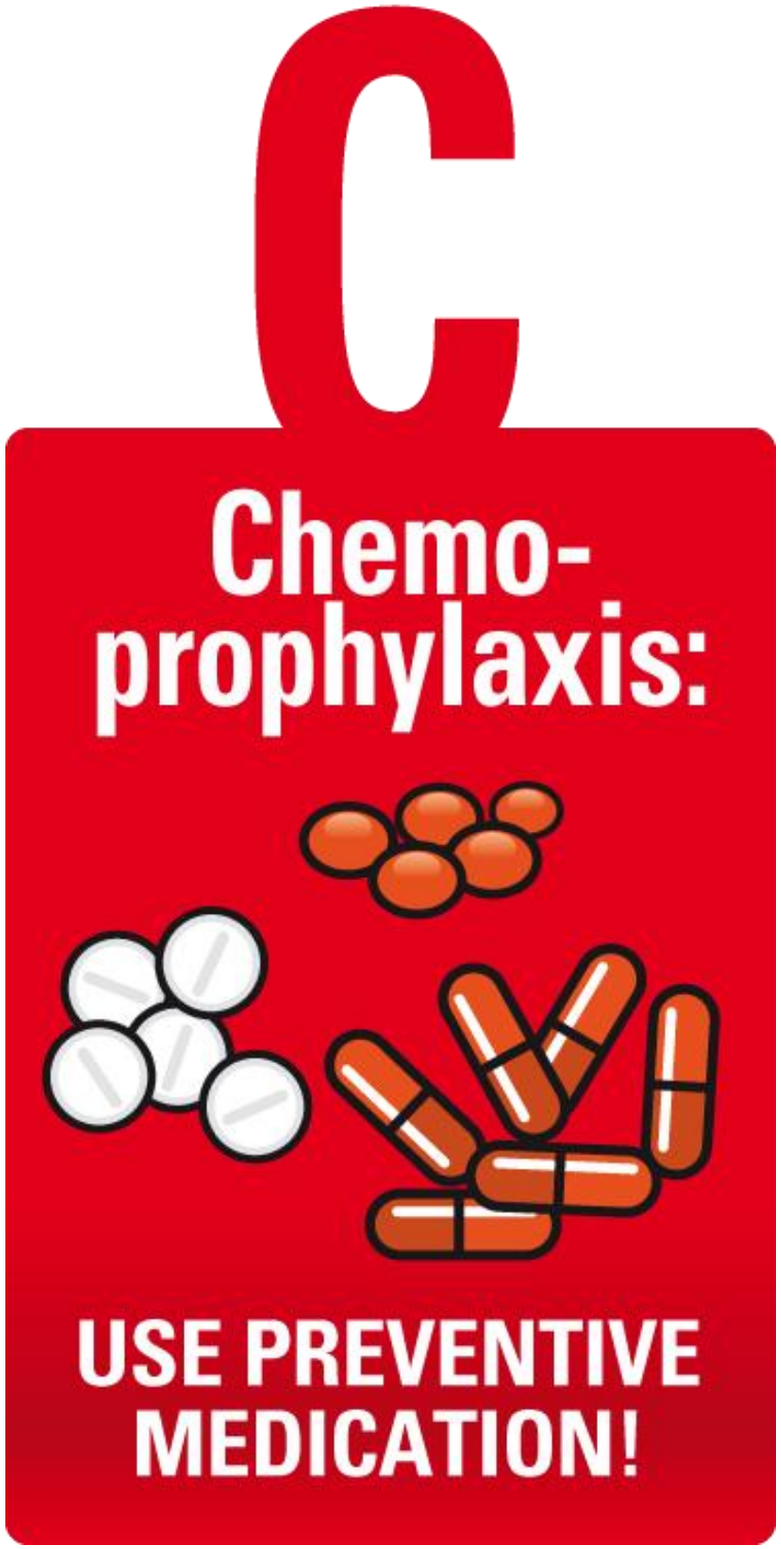
Personal protection (2)

Insecticide–impregnated materials:
nets, curtains, clothing
House screening
House location
Repellents
Fumigants



*** Cost-effective for low-income countries**

Treatment



Take anti-malarial drugs

Mosquito bites cannot be completely avoided.

Regular medication is required to prevent/suppress malaria.

Adequately and promptly treated, malaria is a curable disease.

Chloroquine

(10 mg of base/kg start followed by 5 mg/kg at 12, 24, and 36 h or by 10 mg/kg at 24 h and 5 mg/kg at 48 h)

or

Amodiaquine

(10–12 mg of base/kg qd for 3 days)

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Radical Treatment

Vivax or Ovale

Primaquine

(0.5 mg of base/kg qd) should be given for **14 days** to prevent relapse.

In mild G6PD deficiency, 0.75 mg of base/kg should be given once weekly for 6–8 weeks.

Not be given in severe G6PD deficiency

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Falciparum

Artesunate (3 days)

+

Sulfadoxine/Pyrimethamine as a single dose

or

Artesunate(3 days)

+

Amodiaquine (3 days)

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Multidrug-resistant Falciparum

Artemether-Lumefantrinec

(bid for 3 days with food)

or

Artesunate (3 days)

+

Mefloquine (3 days)

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Second-line treatment/treatment of imported

Artesunate(7 days) or **Quinine** (tid for 7 days)

plus 1 of the following 3:

1. **Tetracycline** (qid for 7 days)

2. **Doxycycline** (qd for 7 days)

3. **Clindamycin** (bid for 7 days)

or

Atovaquone-Proguanil

(qd for 3 days with food)

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Severe Falciparum

Artesunate (**IV** followed by at 12 and 24 h and then daily if necessary)

or, if unavailable, one of the following:

Artemether (**IM** followed by qd)

or

Quinine dihydrochloride (**infused** over 4 h, followed infused over 2–8 h q8h)

or

Quinidine (**infused** over 1–2 h, followed by hour with electrocardiographic monitoring)

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ANTIMALARIAL DRUGS ACCORDING TO WHO

WHO now recommends **Artemisinin**-based combinations as first-line treatment for uncomplicated **Falciparum**

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Treatment of severe malaria

Severe malaria should be treated with injectable artesunate (intramuscular or intravenous) for at least 24 hours and followed by a complete 3-day course of an ACT once the patient can tolerate oral medicines. When injectable treatment cannot be given,

Treatment of *P. vivax* infections

P. vivax infections should be treated with chloroquine in areas where this medicine remains effective. In areas where chloroquine-resistant *P. vivax* has been identified, infections should be treated with an ACT, preferably one in which the partner medicine has a long half-life.

Treatment of *P. falciparum* infections

WHO recommends artemisinin-based combination therapies (ACTs) for the treatment of uncomplicated malaria caused by the *P. falciparum* parasite. By combining 2 active ingredients with different mechanisms of action, ACTs are the most effective antimalarial medicines available today. WHO currently recommends 5 ACTs for use against *P. falciparum* malaria. The choice of ACT should be based on the results of therapeutic efficacy studies against local strains of *P. falciparum* malaria.

Prevention

no safe, effective, long-lasting **vaccine** is likely to be available for general use in the near future

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YOUR ABCD AGAINST MALARIA

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A

Awareness:



RECOGNIZING MALARIA!

B

Bite prevention:



CONTROLLING MOSQUITOES!

C

Chemo-prophylaxis:



USE PREVENTIVE MEDICATION!

D

Diagnosis:



EARLY RECOGNITION!

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See a doctor urgently...

...if you suspect malaria!

- Symptoms may arise **one week or more** after entering a malaria risk area.
- **Immediate treatment** is essential.
- Even if you are taking preventive treatment: **be aware, ensure early diagnosis.**